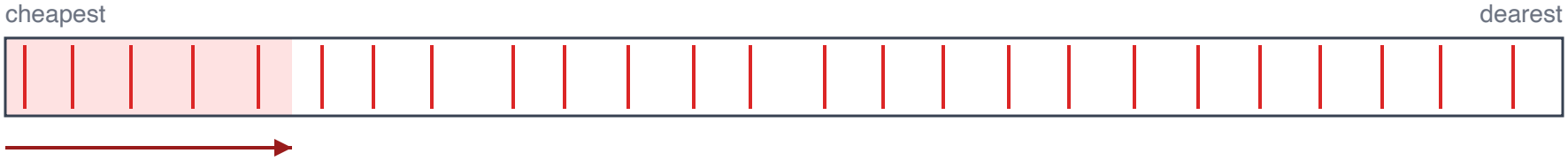
Why a per-row check reads the table for a sparse caller

The Products table has an index on Price. The database can read rows from cheapest to dearest, but it cannot tell which rows the caller may see without testing each one.

Store manager: 100 of 1,000,000 products, spread through the price order



The scan tests rows from the left: 184,205 rows tested before 20 of hers are found (PostgreSQL)

Red marks: her 100 rows. Every other row is tested and rejected.

Far-end caller: 20,000 products, all at the dear end of the price order

980,020 rows tested before the first row of hers appears (PostgreSQL); 44 seconds for one page of 20

Red block: her 20,000 rows, contiguous but at the end. The scan has no way to jump to them.

The planner cannot help.

"Visible to this caller" is not a column. It depends on the caller's grants and the row's place in the tree, so no index on the table makes the visible rows contiguous.